Expectations: Multivariate Random Vector

Let (X_1, X_2) be a random vector and $Y = g(X_1, X_2)$, i.e. a function of the random vector, $\ni \exists\, E(Y)$,

$$E(Y) = \begin{cases} \int\int g(x_1, x_2) \cdot f(x_1, x_2)\, dx_1\, dx_2, & (x_1, x_2) \sim \text{cont R}\overline{\text{V}} \\ \sum\sum g(x_1, x_2) \cdot p(x_1, x_2)\,, & (x_1, x_2) \sim \text{discrete R}\overline{\text{V}} \end{cases}$$

Theorem 2.1.1: Let (X_1, X_2) be a random vector and $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ be random variables whose expectations exist, then $\forall\, k_1, k_2 \in \mathbb{R}$

$$E(k_1 Y_1 + k_2 Y_2) = k_1 E(Y_1) + k_2 E(Y_2)$$

WLOG, Let $X_1, X_2 \sim$ cont. RV

$$\begin{aligned} E(k_1 Y_1 + k_2 Y_2) &= \iint (k_1 Y_1 + k_2 Y_2) \cdot f(x_1, x_2)\, dx_1\, dx_2 \\ &= \iint (k_1 Y_1 f(x_1, x_2) + k_2 Y_2 f(x_1, x_2))\, dx_1\, dx_2 \\ &= \iint k_1 Y_1 f(x_1, x_2)\, dx_1\, dx_2 + \iint k_2 Y_2 f(x_1, x_2)\, dx_1\, dx_2 \\ &= k_1 \iint Y_1 f(x_1, x_2)\, dx_1\, dx_2 + k_2 \iint Y_2 f(x_1, x_2)\, dx_1\, dx_2 \\ &= k_1 E(Y_1) + k_2 E(Y_2) \end{aligned}$$

Logically, for any function $g(X_2)$,

$$E\big(g(X_2)\big) = \int\int g(x_2) \cdot f(x_1, x_2)\, dx_1\, dx_2 = \int g(x_2) \cdot f(x_2)\; dx_2$$

where we integrate out the other random variable, i.e. X_1, and a similar process is used if X_1, X_2 are discrete for any $g(X_1)$, $g(X_2)$, $g(X_1, X_2)$, $g_1(X_1, X_2)$, and/or $g_2(X_1, X_2)$

Example 2.7: Let X_1 and X_2 be defined by the joint PDF

$$f(x_1, x_2) = \begin{cases} 8 \cdot x_1 \cdot x_2, & 0 < x_1 < x_2 < 1 \\ 0, & \text{e/w} \end{cases}$$

Similar to example 2b, we need to be cautious of our bounds, as X_1 is restricted on the UB by X_2 and X_2 is restricted on the LB by X_1

If we define $Y = g(x_1, x_2) = X_1 \cdot X_2^2$, then:

$$E(Y) = E(X_1, X_2^2) = \iint (x_1 x_2^2) \cdot 8 x_1 x_2 \, dx_1 \, dx_2$$

$$= \int_0^1 \int_0^{x_2} 8 x_1^2 x_2^3 \, dx_1 \, dx_2$$

$$= \int_0^1 \left(\frac{8}{3} x_1^3 x_2^3 \right) \Big|_{x_1=0}^{x_1=x_2} dx_2$$

$$= \int_0^1 \frac{8}{3} x_2^3 x_2^3 \, dx_2$$

$$= \int_0^1 \frac{8}{3} x_2^6 \, dx_2$$

$$= \frac{8}{21} x_2^7 \Big|_0^1$$

$$= \frac{8}{21}$$

Exercise: Confirm for yourself that $E(X_2) = 4/5$ and $E(7X_1X_2^2 + 5X_2) = 20/3$.

Definition 2.1.2: Let $\mathbb{X} = (X_1, X_2)'$ be a random vector, then if $E(e^{t_1X_1+t_2X_2})$ $\exists$ for $|\ t_1\ | < h_1$ and $|\ t_2\ | < h_2$, where $h_1, h_2 > 0$, then the MGF of $\mathbb{X}$ is denoted by:

$$M_{X_1,X_2}(t_1, t_2) = M(t_1, t_2) = E(e^{t_1X_1+t_2X_2})$$

Example 2.8: Let X and Y have a joint PDF defined by:

again, mind the bounds

$$f(x,y) = \begin{cases} e^{-y}, & 0 < x < y < \infty \\ 0, & \text{e/w} \end{cases}$$

$$M(t_1, t_2) = E(e^{t_1x+t_2y}) = \iint e^{t_1x+t_2y} e^{-y}\, dy\, dx$$

$$= \iint e^{t_1x} e^{t_2y} e^{-y}\, dy\, dx$$

$$= \iint e^{t_1x} e^{-(1-t_2)y}\, dy\, dx$$

$$= \int_0^\infty \int_x^\infty e^{t_1x} e^{-(1-t_2)y}\, dy\, dx$$

$$= \int_0^\infty e^{t_1x} \int_x^\infty e^{-(1-t_2)y}\, dy\, dx$$

$$= \frac{1}{(1-t_1-t_2)(1-t_2)}$$

assuming $t_2 < 1$, $(t_1+t_2) < 1$

As can be seen in Definition 2.1.2,

$$M_{X_1,X_2}(t_1,t_2) = M(t_1,t_2) = E(e^{t_1X_1+t_2X_2})$$

by evaluating the MGF of the random vector for one of the t_i's at 0, we can find the other marginal MGF. That is, the MGFs of X_1 and X_2 are:

$$M(t_1) = M(t_1, t_2 = 0) = E\left(e^{t_1 x_1 + 0 x_2}\right) = E\left(e^{t_1 x_1}\right)$$

$$M(t_2) = M(t_1 = 0, t_2) = E\left(e^{0 x_1 + t_2 x_2}\right) = E\left(e^{t_2 x_2}\right)$$

Considering the results of example 2.8, the marginal MGF of X and Y are:

$$M(t_1) = M(t_1, t_2 = 0) = \frac{1}{(1-t_1)} = (1-t_1)^{-1}$$

$$M(t_2) = M(t_1 = 0, t_2) = \frac{1}{(1-t_2)(1-t_2)} = (1-t_2)^{-2}$$

if we recall from the univariate examples

$$M(t_1) = (1-t_1)^{-1} \Rightarrow X \sim Exp(\lambda = 1)$$

for X_2, since $f(x) = \int_x^\infty e^{-y}\,dy = -e^{-y}\Big|_x^\infty = e^{-x}$ and for y, we'll leave it to you to prove $f(y) = y \cdot e^{-y}$

this is: $Y \sim \Gamma(\alpha, \beta)$ and $exp(\lambda)$ is a special case of $\Gamma(\alpha, \beta)$

Definition 2.1.3: Let $\mathbb{X} = (X_1, X_2)'$ be a random vector, then if $E(X_1)$ and $E(X_2)$ $\exists$, then the expected value of a random vector exists and is given by:

$$E(\mathbb{X}) = \begin{bmatrix} E(X_1) \\ E(X_2) \end{bmatrix}$$

Bivariate Transformations

Let (X_1, X_2) be a random vector and $Y = g(X_1, X_2)$ be some one-to-one function, to determine the CDF of Y we can perform a similar transformation as we did for the univariate examples.

Discrete: Let $(X_1, X_2) \sim p(x_1, x_2)$ with support $\mathcal{S}$, let $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ be one-to-one transformations such that $\mathcal{S}_X \rightarrow \mathcal{S}_Y$, then

$$p(y_1, y_2) = \begin{cases} p(u_1^{-1}(y_1, y_2), u_2^{-1}(y_1, y_2)), & (y_1, y_2) \in \mathcal{S}_Y \\ 0, & \text{e/w} \end{cases}$$

Example 2.9:

$$p(x_1, x_2) = \frac{\lambda_1^{x_1} \lambda_2^{x_2} e^{-\lambda_1 - \lambda_2}}{x_1!\, x_2!}, \quad x_1, x_2 \in (0, 1, 2, \ldots)$$

let $Y_1 = X_1 + X_2$, even if we are not interested in Y_2, we still need to define it for a bivariate transformation, but we can use any function

let $Y_2 = X_2$, i.e. the simplest transformation

$\Rightarrow X_2 = Y_2 \qquad X_1 = Y_1 - Y_2$

$S_X \rightarrow S_Y = \{(y_1, y_2) : y_1 \in \{0, 1, 2, \ldots\}, y_2 \in \{0, 1, 2, \ldots\}\}$

$$\therefore P_Y(y_1, y_2) = P_X(X_1 = y_1 - y_2, X_2 = y_2)$$
$$= \frac{\lambda_1^{y_1 - y_2} \lambda_2^{y_2} e^{-\lambda_1 - \lambda_2}}{(y_1 - y_2)!\,(y_2)!}$$

Continuous: Let $(X_1, X_2) \sim f_X(x_1, x_2)$ with support $\mathcal{S}$, let $y_1 = u_1(x_1, x_2)$, $y_2 = u_2(x_1, x_2)$, and the Jacobian be defined as:

$$J = \det\left(\left\| \begin{matrix} \dfrac{\partial x_1}{\partial y_1} & \dfrac{\partial x_1}{\partial y_2} \\ \dfrac{\partial x_2}{\partial y_1} & \dfrac{\partial x_2}{\partial y_2} \end{matrix} \right\|\right)$$

then the bivariate transformation can be defined as

$$f_Y(y_1, y_2) = \begin{cases} f_X(u_1^{-1}(y_1, y_2), u_2^{-1}(y_1, y_2)) \cdot |J|, & (y_1, y_2) \in \mathcal{T}_Y \\ 0, & \text{e/w} \end{cases}$$

Example 2.10:

$$f(x_1, x_2) = \begin{cases} 1, & 0 < x_1, x_2 \le 1 \\ 0, & e/w \end{cases}$$

let $Y_1 = X_1 + X_2$, $Y_2 = X_1 - X_2$

$X_1 = Y_1 - X_2$ $\quad X_1 = Y_2 + X_2$

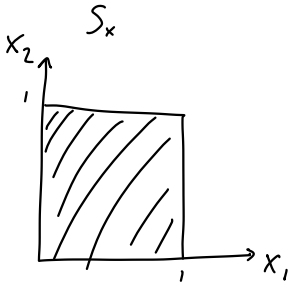
$X_1 = \frac{1}{2}(Y_1 + Y_2)$, $X_2 = \frac{1}{2}(Y_1 - Y_2)$

$$J = \begin{vmatrix} \frac{\partial}{\partial y_1} x_1 & \frac{\partial}{\partial y_2} x_1 \\ \frac{\partial}{\partial y_1} x_2 & \frac{\partial}{\partial y_2} x_2 \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$$

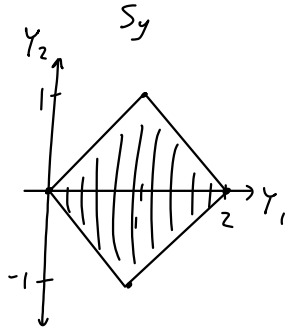
$$\begin{aligned} f_Y(y_1, y_2) &= f_X(x_1 = u_1^{-1}(y_1, y_2), x_2 = u_2^{-1}(y_1, y_2)) \cdot |J| \\ &= 1 \cdot \left|-\tfrac{1}{2}\right| \\ &= \tfrac{1}{2} \end{aligned}$$

	(x_1, x_2)	(y_1, y_2)
$X_1 = \frac{1}{2}(Y_1 + Y_2)$	$(0,0)$	$(0,0)$
$X_2 = \frac{1}{2}(Y_1 - Y_2)$	$(1,0)$	$(1,1)$
$Y_1 = X_1 + X_2$	$(1,1)$	$(2,0)$
$Y_2 = X_1 - X_2$	$(0,1)$	$(1,-1)$

S_x



S_y



The reason the behaviour is not "surprising" is in STAT 333 we saw before that the sum two independent unif(0,1) produces a "triangular distribution" on the support (0,2), which is a special case of the Irwin-Hall distribution, which is the sum of n ind. Unif(0,1); that is, if $X_i \overset{iid}{\sim} \text{Unif}(0,1)$ $\forall\, i \in (1,\dots,n)$ then $\sum_{i=1}^{n} X_i \sim IH(n)$

Given that $Y = u(X_1, X_2)$, or $Y = u_1(x_1, x_2)$ or $Y = u_2(x_1, x_2)$, can be any function, if we let $Y = \exp\,(t \cdot u(X_1, X_2))$, i.e. the MGF, then if we have the following joint PDF

$$f_X(x_1, x_2) = \begin{cases} \frac{1}{4} e^{-\left(\frac{x_1+x_2}{2}\right)}, & 0 < x_1, x_2 < \infty \\ 0, & \text{e/w} \end{cases}$$

We get an interesting transformation for $Y = \frac{1}{2}(X_1 - X_2)$

$$E(e^{tY}) = \iint e^{tY} \cdot f(x_1, x_2)\, dx_1\, dx_2$$

$$= \iint e^{\frac{1}{2}t(x_1 - x_2)} \cdot \frac{1}{4} e^{-\frac{1}{2}(x_1 + x_2)}\, dx_1\, dx_2$$

$$= \iint e^{\frac{t}{2}x_1} e^{-\frac{t}{2}x_2} \underbrace{\frac{1}{2} e^{-\frac{1}{2}x_1} \frac{1}{2} e^{-\frac{1}{2}x_2}}_{\text{both are } \exp(\lambda = \frac{1}{2})}\, dx_1\, dx_2$$

$$= \int_0^\infty \frac{1}{2} e^{-\frac{1}{2}(t+1)x_2} \cdot \left(\frac{1}{1-t}\right) dx_2$$

$$= \left(\frac{1}{1-t}\right)\left(\frac{1}{1+t}\right) = \frac{1}{(1-t^2)}$$

This MGF is for a special distribution $Y \sim Laplace(0,1)$ distribution, also known as the double exponential distribution. Although not obvious, the name double exponential shouldn't be too surprising if we look at the original joint PDF

$$f(x_1, x_2) = \frac{1}{4} e^{-\left(\frac{x_1+x_2}{2}\right)} = \frac{1}{2} e^{-\frac{1}{2}x_1} \cdot \frac{1}{2} e^{-\frac{1}{2}x_2}$$

predictors

number of individuals

The Laplace distribution has real world implications, as it is used in LASSO regression that is commonly employed in data mining when $p \gg n$, i.e. there are more variables than observations. You can learn more in STAT357 and STAT445.

Conditional Distributions and Expectations

Let $X_1, X_2 \sim$ discrete random variable with a joint PMF $p(x_1, x_2)$ and marginal PMFs $p(x_1)$ and $p(x_2)$, then we can define the conditional distribution similar to a conditional probability, whereby

$$P(X_2 = x_2 \,|X_1 = x_1) = \frac{P(X_2 = x_2 \cap X_1 = x_1)}{P(X_1 = x_1)}$$

$$= \frac{p(x_1, x_2)}{p(x_1)} = p(x_2|x_1) \quad \forall\, x_2 \in \mathcal{S}_X, \forall\, x_1 \ni \; p(x_1) > 0$$

Importantly, this is still a probability that must follow all of the previously established rules/laws, i.e.

$$\sum_{x_2} p(x_2|x_1) = \sum_{x_2} \frac{p(x_1,x_2)}{p(x_1)} = \frac{1}{p(x_1)} \sum p(x_1,x_2) = \frac{1}{p(x_1)} \cdot p(x_1) = 1$$

Example 2.11: Using the data in Example 2.3,

		X_2				
		A	B	AB	O	
X_1	⊕	0.33	0.09	0.03	0.37	0.82
	⊖	0.07	0.02	0.01	0.08	0.18
		0.40	0.11	0.04	0.45	

$p(-, A) = 0.07$

$$p(A|-) = \frac{p(-,A)}{p(-)} = \frac{0.07}{0.18}$$

looking at marginal

$$\sum_{x_2} p(-|x_2) = \sum_{x_2} \frac{p(-,x_2)}{p(x_2)} = \frac{0.07}{0.18} + \ldots + \frac{0.08}{0.18} = 1$$

We call $p(x_2 \mid x_1)$ the **conditional PMF of X_2 given $X_1 = x_1$**, where x_1 is just some value that lies within the support of X_1. We can do a similar process for $X_1, X_2 \sim$ continuous random variables with joint PDF $f(x_1, x_2)$

$$\int f(x_2|\, x_1)\; dx_2 = \int \frac{f(x_1, x_2)}{f(x_1)} dx_2$$

$$= \frac{1}{f(x_1)} \int f(x_1, x_2)\, dx_2 = \frac{f(x_1)}{f(x_1)} = 1$$

Importantly, we need to keep in mind that for the PDF, we have to evaluate the probabilities slightly differently because $P(X = x) = 0$ for any continuous random variable. Furthermore, if $u(X_2)$ is some function, the conditional expectation of $u(X_2)$ given $X_1 = x_1$ (assuming it exists) is:

$$E(u(X_2)|X_1 = x_1) = \begin{cases} \sum_{x_2} u(x_2) \cdot p(x_2|x_1)\,, & X_1, X_2 \sim \text{discrete} \\ \int u(x_2) \cdot f(x_2|x_1)\, dx_2, & X_1, X_2 \sim \text{continuous} \end{cases}$$

and, similar to before, when we first introduced the idea of an expectation, the expectation of the value, be it a random variable or function of a random variable, is "inputted" into our function, thus the **conditional mean** is

$$E(X_2|X_1 = x_1) = \begin{cases} \sum_{x_2} x_2 \cdot p(x_2|x_1)\,, & X_1, X_2 \sim \text{discrete} \\ \int x_2 \cdot f(x_2|x_1)\, dx_2, & X_1, X_2 \sim \text{continuous} \end{cases}$$

and the conditional variance is

$$Var(X_2|X_1 = x_1) = E([X_2 - E(X_2|X_1 = x_1)^2]|\, X_1 = x_1)$$

$$= E(X_2^2\, |X_1 = x_1) - E(X_2\, |\, X_1 = x_1)^2$$

It is important to note that we are conditioning on a specific event (i.e. $X_1 = x_1$), even though X_1 is a random variable. **In the context of a conditional statement, it is a known value we condition on**. This is particularly important when you consider which variable you are either summing or integrating over, the random variable of interest (i.e. X_2), so **what we condition on can be thought of as a constant**.

Example 2.12:

$$f(x_1, x_2) = \begin{cases} 6x_2, & 0 < x_2 < x_1 < 1 \\ 0, & e/w \end{cases}$$

$$f(x_1) = \int f(x_1, x_2)\, dx_2 = \int_0^{x_1} 6x_2\, dx_2 = 3x_2^2 \Big|_0^{x_1} = 3x_1^2$$

$$f(x_2) = \int f(x_1, x_2)\, dx_1 = \int_{x_2}^{1} 6x_2\, dx_1 = 6x_2 x_1 \Big|_{x_2}^{1} = 6x_2 - 6x_2^2 = 6x_2(1 - x_2)$$

$$\begin{aligned} E(X_2|X_1 = x_1) &= \int_0^{x_1} x_2 \cdot f(x_2|x_1)\, dx_2 \\ &= \int_0^{x_1} x_2 \cdot \frac{f(x_1, x_2)}{f(x_1)}\, dx_2 \\ &= \int_0^{x_1} x_2 \cdot \frac{6x_2}{3x_1^2}\, dx_2 \\ &= \frac{2}{x_1^2} \int_0^{x_1} x_2^2\, dx_2 \\ &= \frac{2}{3x_1^2} x_2^3 \Big|_0^{x_1} \\ &= \frac{2}{3} x_1 \end{aligned}$$

Since $E(X_2|X_1 = x_1)$ is a function of x_1

$\Rightarrow E(X_2|X_1 = x_1) \sim RV$

$\Rightarrow E(X_2|X_1 = x_1)$ has a μ and σ^2

let $Y = \frac{2}{3}X_1 \Rightarrow X_1 = \frac{3}{2}Y$

Method: Jacobian

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} x \right|$$

$$= 3x_1^2 \Big|_{x_1 = g^{-1}(y)} \cdot \left| \frac{d}{dy} x \right|$$

$$= 3\left(\frac{3}{2}y\right)^2 \cdot \left|\frac{3}{2}\right|$$

$$= \frac{81}{8} y^2$$

where $S_y = \{y : 0 < y < \frac{2}{3}\}$ if $S_{x_1} = \{x_1 : 0 < x_1 < 1\}$

$$E(Y) = \int_0^{2/3} y \cdot \frac{81}{8} y^2 \, dy = \frac{81}{8} \int_0^{2/3} y^3 \, dy = \frac{81}{32} y^4 \Big|_0^{2/3} = \frac{81}{32}\left(\frac{2}{3}\right)^4 = \frac{1}{2}$$

$$\equiv E\left(\frac{2}{3}X_1\right) = \dots = \frac{1}{2}$$

$$Var(Y) = Var\left(\frac{2}{3}X_1\right) = \frac{1}{60}$$

Theorem 2.3.1: Let (X_1, X_2) be a random vector such that $Var(X_2)$ ∃, then

(A) $E(E(X_2 \mid X_1 = x_1)) = E(X_2)$

(B) $Var(E(X_2 \mid X_1 = x_1)) \leq Var(X_2)$

In reality,

$$Var(E(X_2|X_1 = x_1)) + E(Var(X_2|X_1 = x_1)) = Var(X_2)$$

which means that the variances in Theorem 2.3.1 (B) are equal when $E(Var(X_2|X_1 = x_1)) = 0$.

Exercise: Confirm for yourself that,

$Var(X_2|X_1) = E(X_2^2 | X_1 = x_1) - E(X_2|X_1 = x_1)^2 = \frac{1}{18}x_1^2$

$E(Var(X_2|X_1 = x_1)) = E(\frac{1}{18}x_1^2) = \frac{1}{30}$

if true, then by thm 2.12

$\Rightarrow Var(E(X_2|X_1 = x_1)) + E(Var(X_2|X_1 = x_1))$

$= \frac{1}{60} + \frac{1}{30} = \frac{1}{20} = Var(X_2)$

Before we end this section, let's consider the scenario when $E(Var(X_2|X_1 = x_1)) = 0$.

$\Rightarrow$ the "center" of the RV (i.e. $Var(X_2|X_1) = 0$)

$\Rightarrow$ the "average" of the RV is 0

Recall: $Var(X_2|X_1 = x_1) = E(X_2^2 \,|X_1 = x_1) - E(X_2 \mid X_1 = x_1)^2$

If RV $= 0 \Rightarrow E(X_2^2|X_1) - E(X_2|X_1)^2 = 0$

$\Rightarrow E(X_2^2|X_1) = E(X_2|X_1)^2$

If condition holds true $\Rightarrow X_1$ and X_2 are independent